Practice 1

ECON 97: Economic Toolkit

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Question 1

Suppose $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$. What is $P(A \mid B)$?

A. 0.3

B. 0.5

C. 0.6

D. 0.8

Question 2

Suppose $P(A) = 0.4$, $P(B) = 0.7$, and $P(A \cup B) = 0.8$. What is $P(A \cap B)$?

A. 0.1

B. 0.2

C. 0.3

D. 0.4

Question 3

A fair die is rolled twice. What is the probability that the sum is 7?

A. $1/12$

B. $1/9$

C. $1/6$

D. $1/3$

Question 4

A fair die is rolled twice. What is the probability that the maximum number rolled is at most 4?

A. $1/9$

B. $1/4$

C. $4/9$

D. $2/3$

Question 5

Let $X \sim \text{Binomial}(10, 0.3)$. What is $E[X]$?

A. 0.3

B. 3

C. 7

D. 10

Question 6

Let $X \sim$ Binomial$(10, 0.3)$. What is Var(X)?

A. 0.21

B. 2.1

C. 3

D. 7

Question 7

Let $X \sim$ Poisson(5). What is Var(X)?

A. 1

B. $\sqrt{5}$

C. 5

D. 25

Question 8

Let $X \sim \text{Poisson}(3)$. What is $P(X = 0)$?

A. e^{-3}

B. $3e^{-3}$

C. $1 - e^{-3}$

D. e^{3}

Question 9

Let X have pdf

$$f(x) = cx, \qquad 0 \leq x \leq 2.$$

What is c?

A. $1/4$

B. $1/2$

C. 1

D. 2

Question 10

For the same random variable, what is $P(X \leq 1)$?

A. $1/4$

B. $1/2$

C. $3/4$

D. 1

Question 11

Let X have cdf

$$F(x) = x^2, \qquad 0 \leq x \leq 1.$$

What is $P(X > 0.5)$?

A. 0.25

B. 0.50

C. 0.75

D. 1.00

Question 12

Suppose $X \sim N(20, 9)$. What is $P(X \leq 20)$?

A. 0.16

B. 0.50

C. 0.84

D. 0.95

Question 13

Suppose $X \sim N(20, 9)$. What is $P(X > 23)$ approximately?

A. 0.16

B. 0.34

C. 0.50

D. 0.84

Question 14

Suppose $X_1, \ldots, X_{100}$ are i.i.d. with mean $\mu = 50$ and variance $\sigma^2 = 100$. What is the standard deviation of $\bar{X}$?

A. 1

B. 10

C. 100

D. 1000

Question 15

Suppose $X_1, \ldots, X_{100}$ are i.i.d. with mean $\mu = 50$ and variance $\sigma^2 = 100$. Using the CLT, what is

$$P(\bar{X} > 52)$$

approximately?

A. 0.023

B. 0.159

C. 0.500

D. 0.841

Question 16

Suppose $X_1, \ldots, X_{64}$ are i.i.d. with mean $\mu = 10$ and variance $\sigma^2 = 16$. Using the CLT, what is

$$P(\bar{X} < 9)$$

approximately?

A. 0.023

B. 0.159

C. 0.500

D. 0.841

Question 17

Suppose X has mean 100 and variance 25. Using Chebyshev's inequality, what upper bound can we place on

$$P(|X - 100| \geq 10)?$$

A. 0.05

B. 0.10

C. 0.25

D. 0.50

Question 18

Suppose X has mean 8 and variance 4. Using Chebyshev's inequality, what lower bound can we place on

$$P(4 < X < 12)?$$

A. 0.50

B. 0.75

C. 0.875

D. 0.95

Answers

1: C
2: C
3: C
4: C
5: B
6: B
7: C
8: A
9: B
10: A
11: C
12: B
13: A
14: A
15: A
16: A
17: C
18: B